

WiFi Training Program – Module 1 (Final Assignment Answers)

1. OSI Layer:

Wi-Fi operates in Physical (Layer 1) and Data Link (Layer 2).

2. Devices & Properties:

Smartphone, Laptop

Frequency: 2.4/5 GHz

Signal: -45 dBm

Speed: 72–433 Mbps

Security: WPA2/WPA3

3. BSS & ESS:

BSS: Single AP

ESS: Multiple APs with same SSID

4. Access Point Functions:

Connectivity, authentication, traffic management, SSID broadcast

Feature	Bridge Mode	Repeater Mode
Purpose	Connect networks	Extend coverage
Speed	High	Reduced

6. 802.11a vs 802.11b:

5 GHz vs 2.4 GHz, 54 Mbps vs 11 Mbps

7. 2.4 vs 5 GHz:

2.4 GHz → Range

5 GHz → Speed

8. IEEE vs WFA:

IEEE defines standards, WFA certifies devices

9. Backhaul:

Wired (Ethernet/Fiber), Wireless (Mesh)

10. Topologies:

Infrastructure, Ad-hoc, Mesh